

CS231n Deep Learning for Computer Vision

Course Website

Assignment 3

This assignment is due on **Friday, May 30 2025** at 11:59pm PST.

Starter code containing Colab notebooks can be [downloaded here](#).

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Setup

Please familiarize yourself with the [recommended workflow](#) before starting the assignment. You should also watch the Colab walkthrough tutorial below.



CS231N Google Colab Assignment Workflow Tutorial

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Watch on

Note. Ensure you are periodically saving your notebook (**File -> Save**) so that you don't lose your progress if you step away from the assignment and the Colab VM disconnects.

While we don't officially support local development, we've added a **requirements.txt** file that you can use to setup a virtual env.

Once you have completed all Colab notebooks **except** `collect_submission.ipynb`, proceed to the [submission instructions](#).

Goals

In this assignment, you will implement language networks and apply them to image captioning on the COCO dataset. Then you will be introduced to self-supervised learning to automatically learn the visual representations of an unlabeled dataset. Next, you will implement diffusion models (DDPMs) and apply them to image generation. Finally, you will explore CLIP and DINO, two self-supervised learning methods that leverage large amounts of unlabeled data to learn visual representations.

The goals of this assignment are as follows:

- Understand and implement Transformer networks. Combine them with CNN networks for image captioning.
- Understand how to leverage self-supervised learning techniques to help with image classification tasks.
- Implement and understand diffusion models (DDPMs) and apply them to image generation.
- Implement and understand CLIP and DINO, two self-supervised learning methods that leverage large amounts of unlabeled data to learn visual representations.

You will use PyTorch for the majority of this homework.

Q1: Image Captioning with Transformers

The notebook `Transformer_Captioning.ipynb` will walk you through the implementation of a Transformer model and apply it to image captioning on COCO.

Q2: Self-Supervised Learning for Image Classification

In the notebook `Self_Supervised_Learning.ipynb`, you will learn how to leverage self-supervised pretraining to obtain better performance on image classification tasks. **When first opening the notebook, go to** `Runtime > Change runtime type` **and set** `Hardware accelerator` **to** `GPU`.

Q3: Denoising Diffusion Probabilistic Models

In the notebook `DDPM.ipynb`, you will implement a Denoising Diffusion Probabilistic Model (DDPM) and apply it to image generation.

Q4: CLIP and Dino

In the notebook `CLIP_DINO.ipynb`, you will implement CLIP and DINO, two self-supervised learning methods that leverage large amounts of unlabeled data to learn visual representations.

Submitting your work

Important. Please make sure that the submitted notebooks have been run and the cell outputs are visible.

Once you have completed all notebooks and filled out the necessary code, you need to follow the below instructions to submit your work:

1. Open `collect_submission.ipynb` in Colab and execute the notebook cells.

This notebook/script will:

- Generate a zip file of your code (`.py` and `.ipynb`) called `a3_code_submission.zip`.
- Convert all notebooks into a single PDF file called `a3_inline_submission.pdf`.

If your submission for this step was successful, you should see the following display message:

```
###      Done!      Please      submit      a3_code_submission.zip      and  
a3_inline_submission.pdf to Gradescope. ###
```

2. Submit the PDF and the zip file to Gradescope.

Remember to download `a3_code_submission.zip` and `a3_inline_submission.pdf` locally before submitting to Gradescope.

